

Affix ordering in Seediq and other Formosan languages

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main claims:

- the alternation between actor voice (AV) prefix *m-* and infix *<m>* is phonologically conditioned
 - apparent alternation from agentivity/transitivity is due to the presence of an additional vowel in anti-causative (or middle voice) contexts, which spells-out a little *v* head
 - the vowel is obscured by phonological reduction of pre-stress vowels in Seediq
- the order of affix insertion in Formosan is: $\text{Asp}_{\text{PFV}} > \text{voice} > \text{Asp}_{\text{RED}} > v$
 - n.b. this is based off of the ordering of left-aligned voice affixes such as AV and IV

1 basic Seediq (and Formosan) facts

- voice and aspect morphology reconstructs to Proto-Austronesian (PAN):
 - cf. Blust 2013, Chen 2017, Begus et al. 2023 for voice vs. nominalizer functions
 - actor voice (AV), patient voice (PV), locative voice (LV), and instrumental voice (IV)
 - perfective aspect or past tense **<in>*; imperfective aspect or future tense **RED_{CV}-*
 - the functions of these affixes has shifted or been loss in some Formosan languages
- see appendix for verbal morphology paradigms of different Formosan languages

(1) voice alternations (Paiwan; Ferrell 1979)¹

- a. q<**mau] tua vavuy i gadu tua vulu
 <AV>hunt PIV man OBL pig LOC mountain OBL spear**
- b. qalup-**en** nua caucau [a vavuy] i gadu tua vulu
 hunt-PV GEN man PIV pig LOC mountain OBL spear
- c. qalup-**an** nua caucau tua vavuy [a gadu] tua vulu
 hunt-LV GEN man OBL pig PIV mountain OBL spear
- d. **si**-qalup nua caucau tua vavuy i gadu [a vulu]
 IV-hunt GEN man OBL pig LOC mountain PIV spear
 ‘The man hunts wild pigs in the mountains with a spear.’

- modern Seediq has the four voices, but some functions have been reduced/shifted
 - AV and PV alternations are still productive, for the promotion of the agent vs. patient
 - PV and LV are both used for patient promotion, with different levels of affectedness²
 - IV is used for instruments, benefactors, etc. but not productive with TAM morphology

- (2) a. wada **m**-ita arung [ka Habo]
 PST AV-see pangolin PIV Habo
 ‘Habo saw a pangolin.’ actor voice (AV)
- b. wada qta-**un** na Habo [ka arung]
 PST see-PV GEN Habo PIV pangolin
 ‘Habo saw the pangolin.’ patient voice (PV)

¹the glossing used in this handout is PIV = ‘pivot’ and GEN = ‘genitive’ for non-pivot agents and subjects

²the locative use of LV is rare in verbal contexts and more often used in nominalized contexts (Aldridge 2018)

- Seediq has undergone neutralization of all pre-stress vowels (Yang 1976)
 - vowels in pre-penultimate syllables are phonologically reduced to an [u] or [ə]
 - exception #1: pre-stress vowels will match the following vowel if followed by /ʔ/
 - exception #2: pre-stress vowels are elided if preceded by a non-initial /m/ or /n/
- in the orthography, all pre-stress vowels are omitted unless followed by a glottal stop

- (3) a. *tgdaya* ‘up the hill’ → [tugu-daya] d. *seediq* ‘person’ → [se-ʔediq]
 b. *smipaq* ‘kill’ → [su-mipaq] e. *kntetetu* ‘often’ → [kunte-ʔetu]
 c. *qbheni* ‘bird’ → [qubu-heni] f. *smkuxul* ‘like’ → [sum-kuxul]

- the full paradigm for Seediq verbal morphology (adapted from Holmer 1999)
 - a lot of morphological behavior here is reconstructable to PAn (Blust 2013)
 - voice is mostly compositional with tense/aspect but has portmanteau forms with mood
 - reduplication is used in prospective aspect and future tense (cf. incomplete in Tagalog)

(4)

	AV	PV	LV	IV
PRES	m- or ⟨m⟩	-un	-an	s-
PFV	m-n- or ⟨m-n⟩	⟨n⟩	⟨n⟩ -an	-
PSV	m-RED- or RED⟨m⟩-	RED- -un	RED- -an	-
FUT	m-p-	RED- -un	RED- -an	-
IMP	∅	-i	-an-i	-
SUBJ	m- -a or ?	-o	-e	-

2 prefixation vs. infixation

- at first glance, the distribution of AV prefix vs. infix is sensitive to syntactic information:
 - agentive verbs (e.g. unergatives and transitives) take as a baseline the infix ⟨m⟩
 - non-agentive verbs (e.g. unaccusative and statives) take as a baseline the prefix *m-*
 - however, the infix ⟨m⟩ will surface as the prefix *m-* with vowel-initial roots³
- this insight has also been described in terms of verb classes in PAn (Ross 2015)

- (5) surface distribution of prefix vs. infix AV:

- a. C-initial *and* agentive roots → ⟨m⟩
 b. elsewhere → *m-*
 c. idiosyncratic (often roots with three or more syllables) → ∅

- (6) a. unaccusative: *m-rkerak* ‘itch’, *m-sngari* ‘remain’, *m-huqil* ‘die’, *m-seyaq* ‘cough’, *m-takur* ‘fall’, *m-eeniq* ‘exist’, *m-iicuw* ‘be scared’, *dehuk* ‘arrive’, *bsukan* ‘be drunk’
 b. unergative: *s⟨m⟩rebu* ‘poop’, *t⟨m⟩alang* ‘run’, *r⟨m⟩engo* ‘talk’, *l⟨m⟩ingis* ‘cry’, *k⟨m⟩eeki* ‘dance’, *m-uuyas* ‘sing’, *m-eyah* ‘come’, *m-alix* ‘give up’, *skiya* ‘fly’, *qduriq* ‘flee’
 c. transitive: *g⟨m⟩beling* ‘dig’, *c⟨m⟩hungi* ‘forget’, *s⟨m⟩ipaq* ‘kill’, *k⟨m⟩eruc* ‘cut’, *c⟨m⟩ebu* ‘shoot’, *d⟨m⟩urun* ‘ask’, *m-ekan* ‘eat’, *m-angal* ‘take’, *m-etur* ‘kick’, *hpure* ‘cook’

³it is possible that vowel-initial roots are really /ʔ/-initial, and there is no phonologically conditioned prefixation

- this distinction is not tied to lexical items, as some roots can alternate with both allomorphs
 - transitive forms vs. their reflexive/reciprocal/detransitive/anticausative counterparts
- (7) some roots can alternate with both prefix and infix (Holmer 1999, Chen 2018)
- | | |
|--|---|
| <p>a. $t\langle m \rangle$utuy=ku heya
 $\langle AV \rangle$wake=1SG.PIV 3.SG
 ‘I am waking her up.’</p> <p>b. m-tutuy laqi=su
 AV-wake child=2SG.ERG
 ‘Your child is getting up.’</p> | <p>c. wada d$\langle m \rangle$engu qhuni ka Dakis
 PST $\langle AV \rangle$roast wood PIV Dakis
 ‘Dakis heated the wood.’</p> <p>d. wada m-dengu ka qhuni
 PST AV-roast PIV wood
 ‘The wood has been heated.’</p> |
|--|---|
- prefix *m*- has been attributed to several historically independent affixes in PAn (Chen 2023)
 - actor voice **m*-, stative **ma*-, detransitive **mu*-, and reciprocal **maCa*-
 - I argue that at least in Seediq (and potentially PAn) these affixes are decomposable
- (8)
- | | |
|---|--|
| <p>a. actor voice: m-<i>ekan</i> ‘eat’, m-<i>osa</i> ‘go’</p> <p>b. stative: m-<i>narux</i> ‘be sick’, m-<i>tanah</i> ‘be red’</p> <p>c. detransitive: m-<i>takur</i> ‘fall’, m-<i>gerung</i> ‘be broken’</p> <p>d. reciprocal: m-<i>g-gaalu</i> ‘care for each other’</p> | <p>(vs. C-initial $t\langle m \rangle$<i>alang</i> ‘run’)</p> <p>(vs. transitive $t\langle m \rangle$<i>akur</i> ‘trip’)</p> <p>(vs. transitive $g\langle m \rangle$<i>aalu</i> ‘care’)</p> |
|---|--|

proposal: non-agentive (anti-causative?) *v* head is a vowel that prefixes to roots before AV

- cf. Chen 2020 for decomposing PAn **mu*- into actor voice **m*- and detransitive **u*-
 - its distribution is similar to middle voice (Kemmer 1993): reflexives and reciprocals, statives and inchoatives, experiencer and psych verbs, deadjectival and denominal verbs
 - this unifies 1) the prefix vs. infix alternation and 2) the multiple homophonous *m*- affixes
 - the former is a purely phonological alternation between V-initial and C-initial *stems*⁴
 - the latter is decomposed into AV allomorph *m*- and additional morphosyntactic material
 - explains why vowel-initial non-agentive verbs always seem to occur with an extra vowel
- (9) prefixed vowel in non-agentive verbs: AV always surfaces as a prefix
- | |
|--|
| <p>a. <i>rkerak</i> ‘itch’ → V-<i>rkerak</i> ‘itch.v_{ANTICAUS}’ → m-V-<i>rkerak</i> ‘itch.AV.v_{ANTICAUS}’ → <i>mrkerak</i></p> <p>b. <i>eniq</i> ‘exist’ → V-<i>eniq</i> ‘exist.v_{ANTICAUS}’ → m-V-<i>eniq</i> ‘exist.AV.v_{ANTICAUS}’ → <i>meeniq</i></p> <p>c. <i>icuw</i> ‘scared’ → V-<i>icuw</i> ‘scared.v_{ANTICAUS}’ → m-V-<i>icuw</i> ‘scared.AV.v_{ANTICAUS}’ → <i>miicuw</i></p> |
|--|
- (10) no prefixed vowel in agentive verbs: AV surfaces as either a prefix or an infix
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|--|
| <p>a. <i>talang</i> ‘run’ → <i>talang</i> ‘run.v_{AG}’ → $t\langle m \rangle$<i>alang</i> ‘run.AV.v_{AG}’ → <i>tmalang</i></p> <p>b. <i>osa</i> ‘go’ → <i>osa</i> ‘go.v_{AG}’ → m-<i>osa</i> ‘go.AV.v_{AG}’ → <i>mosa</i></p> <p>c. <i>chungi</i> ‘forget’ → <i>chungi</i> ‘forget.v_{AG}’ → $c\langle m \rangle$<i>hungi</i> ‘forget.AV.v_{AG}’ → <i>cmhungi</i></p> |
|--|
- (11)
- | | |
|---|---|
| <p>a. <i>m-ekan</i> ‘AV-eat’</p> <p>b. m-V-<i>narux</i> ‘AV-STAT-be.sick’</p> | <p>c. m-V-<i>takur</i> ‘AV-DETR-fall’</p> <p>d. m-V-<i>g-gaalu</i> ‘AV-REC-care.for.e.o.’</p> |
|---|---|

⁴it is possible that this alternation is instead between AV attaching to derived stems vs. roots (p.c. Lisa Travis)

- this merger in Seediq is aided by the neutralization of pre-stress vowels
 - because roots are at a minimum bisyllabic, this additional vowel will always be located in a pre-stress syllable and be phonologically reduced, obscuring its underlying nature
- there are three morphemes that spell-out the little *v* head: v_{ANTICAUS} , v_{AG} , and v_{CAUS}
 - there are other cases where *v* is overt, such as v_{CAUS} that is realized as *p-* (PAn **pa-*)
 - AV is infixated underlyingly, but does not surface since /m/ is not allowed after labials
 - when /m/ follows a labial, it deletes the labial unless it is a contentful morpheme like *p-*

(12) prefixed *p-* in causativized verbs: AV does not surface

- a. *talang* ‘run’ → ***p-talang*** ‘run. v_{CAUS} ’ → ***p$\langle m \rangle$-talang*** ‘run.AV. v_{CAUS} ’ → *ptalang*
- b. *osa* ‘go’ → ***p-osa*** ‘go. v_{CAUS} ’ → ***p$\langle m \rangle$-osa*** ‘go.AV. v_{CAUS} ’ → *posa*
- c. *chungi* ‘forget’ → ***p-chungi*** ‘forget. v_{CAUS} ’ → ***p$\langle m \rangle$-chungi*** ‘forget.AV. v_{CAUS} ’ → *pchungi*

- (13) a. *p$\langle m \rangle$atis* ‘$\langle \text{AV} \rangle$write’ → [matis] c. *q$\langle m \rangle$ita* ‘$\langle \text{AV} \rangle$see’ → [mita]
 b. *b$\langle m \rangle$ege* ‘$\langle \text{AV} \rangle$buy’ → [mege] d. *k$\langle m \rangle$hemuc* ‘$\langle \text{AV} \rangle$at.will’ → [muhemuc]

- all that is needed to account for the data are the following vocabulary insertion rules:⁵
 - the exact location where the AV infix $\langle m \rangle$ is inserted will be discussed later

- (14) a. AV ⇔ *m-* / $_V$ or $\langle m \rangle$ / $_C$
 b. v_{ANTICAUS} ⇔ V- (with additional reduplication for reciprocals)
 c. v_{AG} ⇔ $\emptyset$
 d. v_{CAUS} ⇔ *p-*

3 affix ordering

proposal: the order of verbal affixes in Formosan languages is: $\text{Asp}_{\text{PFV}} > \text{voice} > \text{Asp}_{\text{RED}} > v$

- n.b. this is based off of the ordering of left-aligned voice affixes such as AV and IV
- this is potentially at odds with theories where voice morphology is high (Chen 2017, 2023)

3.1 voice and *v*

- voice is ordered after the little *v* head, which includes v_{ANTICAUS} and v_{CAUS}
 - otherwise, AV would infix into C-initial roots regardless of the presence of an overt *v*
 - and AV would prefix onto V-initial roots directly without the extra vowel

- (15)
- | | | |
|----------------------------|--|---|
| | voice > v_{ANTICAUS} | * v_{ANTICAUS} > voice |
| a. <i>sngari</i> ‘remain’: | <i>m- V-sngari</i> [musu-ngari] | * <i>V-s<math>\langle m \rangle</math>ngari</i> [(u)sum-ngari] |
| b. <i>huqil</i> ‘die’: | <i>m- V-huqil</i> [mu-huqil] | * <i>V-h<math>\langle m \rangle</math>uqil</i> [(u)hu-muqil] |
| c. <i>uwic</i> ‘be tired’: | <i>m- V-uwic</i> [mu-ʔuwic] | * <i>V-m-uwic</i> [(u)-muwic] |

⁵there is potentially also a stative *k-* that is not discussed here, which does not show up in basic AV forms but does in cases where there is no overt AV marker, such as negative AV forms or PV and LV forms; this requires further investigation, but one possibility is that /k/ is always present but deleted via a similar phonological process as (13); it is unclear whether this stative *k-* should be v_{ANTICAUS} or its own vocabulary item v_{STAT} , more data is needed here

(16)		voice > v_{CAUS}	$*v_{\text{CAUS}} > \text{voice}$
a.	<i>talang</i> ‘run’:	$p\langle m \rangle\text{-talang}$ [pu-talang]	$*p\text{-}t\langle m \rangle\text{alang}$ [putu-malang]
b.	<i>osa</i> ‘go’:	$p\langle m \rangle\text{-osa}$ [posa]	$p\text{-}m\text{-osa}$ [posa]
c.	<i>chungi</i> ‘forget’:	$p\langle m \rangle\text{-chungi}$ [pucu-hungi]	$*p\text{-}c\langle m \rangle\text{hungi}$ [pucum-hungi]

3.2 inner aspect

- a low Aspect projection has been proposed for Austronesian languages (Travis 1991, 2010)
 - this projection sits below Voice, and is realized through CV-reduplication
 - reflects internal structure of an event, namely some kind of incomplete/ongoing action

(17)	a.	<i>ciga</i>	<i>m-e~ekan</i> =[ku]	<i>ido</i>	<i>m-eyah</i>	Pawan	
		yesterday	AV-RED~eat=1SG.PIV	rice	AV-come	Pawan	
		‘Yesterday I was about to eat when Pawan came.’					prospective [Seediq; fieldwork]
	b.	$d\langle em \rangle a\sim deru$	[i Atrung]	<i>dra</i>	<i>patraka</i>		
		$\langle AV \rangle \text{RED} \sim \text{cook}$	PIV Atrung	ACC	meat		
		‘Atrung is cooking meat.’					progressive [Puyuma; Chen 2023]
	c.	$s\langle em \rangle iu\sim siup$	[ti Zepul]	<i>nu</i>	$s\langle em \rangle iaw$		
		$\langle AV \rangle \text{RED} \sim \text{suck}$	PIV Zepul	TEMP	$\langle AV \rangle \text{soup}$		
		‘Zepul sucks (it) when she eats soup.’					habitual [Paiwan; Chang 2006]

- on the one hand, it appears that voice is ordered *after* Asp_{RED} with C-initial roots
 - AV infixes into the reduplicated material instead of the verb root (see above)
 - needs testing: IV would prefix onto the reduplicated material instead of the verb root
 - an assumption here is that in Seediq, reduplicated material in over three syllables from the right edge is deleted (Holmer 1999), thus the AV infix surfaces as a prefix
 - deletion is allowed because the result is not ambiguous with AV infixing into the root

(18)		voice > Asp_{RED}	$\text{Asp}_{\text{RED}} > \text{voice}$
a.	<i>?ekan</i> ‘eat’:	$? \langle m \rangle \sim ?ekan$ [me-?ekan]	$*? \sim ? \langle m \rangle ekan$ [(?u?u)-mekan]
b.	<i>rengo</i> ‘talk’:	$r \langle m \rangle \sim rengo$ [mu-rengo]	$*r \sim r \langle m \rangle engo$ [(ru)ru-mengo]
c.	<i>gbeling</i> ‘dig’:	$g \langle m \rangle \sim gbeling$ [mugu-beling]	$*g \sim g \langle m \rangle beling$ [(gu)gumu-beling]

- on the other hand, it appears that voice is ordered *before* Asp_{RED} with V-initial roots

(19)		$*\text{voice} > \text{Asp}_{\text{RED}}$	$\text{Asp}_{\text{RED}} > \text{voice}$
a.	<i>osa</i> ‘go’:	$*m\text{-}o \sim osa$ [mo-osa]	$m \sim m\text{-}osa$ [mu-mosa]
b.	<i>aha</i> ‘go’:	$*m\text{-}a \sim aha$ [ma-aha]	$m \sim m\text{-}aha$ [mu-maha]
c.	<i>eyah</i> ‘come’:	$*m\text{-}e \sim eyah$ [me-eyah]	$m \sim m\text{-}eyah$ [mu-meyah]

- some possible explanations (?):
 1. the prospective affix itself is a prefix *m-* and not reduplication (Holmer 1999), which captures the facts if $\text{voice} > \text{Asp}_{\text{RED}}$, but this does not extend to PV and LV forms
 2. an OT (instead of derivational) approach to morphology?
 3. the forms in (19) are a repair because reduplication wants a consonant (CV sequence)

- (20) a. *usa* ‘go’ → *C-usa* ‘go.Asp_{RED}’ → *C⟨m⟩-usa* ‘go.AV.Asp_{RED}’ → *mmusa*
 b. *aha* ‘go’ → *C-aha* ‘go.Asp_{RED}’ → *C⟨m⟩-aha* ‘go.AV.Asp_{RED}’ → *mmaha*
 c. *eyah* ‘come’ → *C-eyah* ‘come.Asp_{RED}’ → *C⟨m⟩-eyah* ‘come.AV.Asp_{RED}’ → *mmeyah*

- | | | | |
|------|----------------------|---|---|
| (21) | | $\text{Asp}_{\text{RED}} > v$ | $*v > \text{Asp}_{\text{RED}}$ |
| a. | <i>heya</i> ‘stand’: | $\mathbf{p}\langle \mathbf{m} \rangle \sim p\text{-heyu}$ [pupu-heyu] | $*\mathbf{p}\langle \mathbf{m} \rangle \sim \mathbf{h}\text{-heyu}$ [puhu-heyu] |
| b. | <i>osa</i> ‘go’: | $\mathbf{p}\langle \mathbf{m} \rangle \sim p\text{-osa}$ [pu-posa] | $*\mathbf{p}\langle \mathbf{m} \rangle \sim \mathbf{o}\text{-osa}$ [po-osa] |
| c. | <i>talang</i> ‘run’: | $\mathbf{p}\langle \mathbf{m} \rangle \sim p\text{-talang}$ [pupu-talang] | $*\mathbf{p}\langle \mathbf{m} \rangle \sim \mathbf{t}\text{-talang}$ [putu-talang] |

- perfective aspect is well attested in Austronesian languages (Blust 2013)
 - many Formosan languages have lost the infix $*\langle in \rangle$ and instead have a prefix or enclitic that attaches outside of voice (and sometimes even outside of pronominal clitics)
 - three languages preserve the perfective infix: Bunun, Paiwan, and Seediq
 - PV.PFV forms surface as just the PFV infix, which is reconstructable to PAN

- a. tu=idap-**ay**=**lra** [na sawka] ni Risem
3SG.GEN=sweep-LV=PFV PIV kitchen GEN Risem
'Risem swept the house.'
[Puyuma; Zheng 2018]
- b. kuna-**an**=na=**ti** [ya sunis]
first-PV=3SG.GEN=PFV PIV child
'He surpassed the child.'
[Kavalan; Chang 2006]

- (23) a. Bunun (NTU 2016): **m**⟨**i**⟩-ludah ‘AV⟨PFV⟩-hit’ (vs. l⟨**in**⟩udah ‘⟨PV.PFV⟩hit’)
s⟨**in**⟩-pit’ia ‘IV⟨PFV⟩-cook’ (vs. p⟨**in**⟩it’ia ‘⟨PV.PFV⟩cook’)
b. Paiwan (Chang 2018): **na**-t⟨**em**⟩ekel ‘PFV-⟨AV⟩drink’ **na**-k⟨**em**⟩an ‘PFV-⟨AV⟩eat’
s⟨**in**⟩**i**-kelem ‘IV⟨PFV⟩-hit’ **s**⟨**in**⟩**i**-patjavat ‘IV⟨PFV⟩-leave’
c. Seediq (fieldwork): **m**⟨**n**⟩-ekan ‘AV⟨PFV⟩-eat’ **m**⟨**n**⟩-imah ‘AV⟨PFV⟩-drink’
r⟨**m-n**⟩engo ‘⟨AV-PFV⟩talk’ l⟨**m-n**⟩ingis ‘⟨AV-PFV⟩cry’

(24)		$\text{Asp}_{\text{PFV}} > \text{voice} / _V$	$*\text{voice} > \text{Asp}_{\text{PFV}} / _V$
		$\text{voice} > \text{Asp}_{\text{PFV}} / C_$	$*\text{Asp}_{\text{PFV}} > \text{voice} / C_$
	a.	<i>rengo</i> ‘talk’: $r\langle \mathbf{m-n} \rangle \text{engo}$ [rum-nengo]	$*r\langle \mathbf{n-m} \rangle \text{engo}$ [run-mengo]
	b.	<i>osa</i> ‘go’: $\mathbf{m}\langle \mathbf{n} \rangle\text{-osa}$ [mu-nosa]	$*\mathbf{n}\langle \mathbf{m} \rangle\text{-osa}$ [nu-mosa]
	c.	<i>chungi</i> ‘forget’: $c\langle \mathbf{m-n} \rangle \text{hungi}$ [cumun-hungi]	$*c\langle \mathbf{n-m} \rangle \text{hungi}$ [cunum-hungi]

- I adopt the $_V$ environment, for consistency with other Formosan languages

(25)	a.	<i>rengo</i> ‘talk’ $\rightarrow r\langle \mathbf{m} \rangle \text{engo}$ ‘talk.AV’ $\rightarrow r\langle \mathbf{m-n} \rangle \text{engo}$ ‘talk.AV.PFV’ $\rightarrow \text{rmnengo}$
	b.	<i>osa</i> ‘go’ $\rightarrow \mathbf{m}\text{-osa}$ ‘go.AV’ $\rightarrow \mathbf{m}\langle \mathbf{n} \rangle\text{-osa}$ ‘go.AV.PFV’ $\rightarrow \text{mnusa}$
	c.	<i>chungi</i> ‘forget’ $\rightarrow c\langle \mathbf{m} \rangle \text{hungi}$ ‘talk.AV’ $\rightarrow c\langle \mathbf{m-n} \rangle \text{hungi}$ ‘talk.AV.PFV’ $\rightarrow \text{cmnhungi}$

4 summary

(26)	a.	AV	$\Leftrightarrow m- / _V$	
			$\langle m \rangle / _C$	(infix location is $_V$, including pre-stress vowels)
	b.	v_{ANTICAUS}	$\Leftrightarrow V-$	
	c.	v_{AG}	$\Leftrightarrow \emptyset$	
	d.	v_{CAUS}	$\Leftrightarrow p-$	
	e.	RED	$\Leftrightarrow C-$	
	f.	PFV	$\Leftrightarrow \langle n \rangle$	(infix location is $_V$, excluding pre-stress vowels)

- the alternation between AV prefix $m-$ and infix $\langle m \rangle$ is phonologically conditioned
 - there are three morphemes that spell-out the little v head: v_{ANTICAUS} , v_{AG} , and v_{CAUS}
- the order of affix insertion in Formosan is: $\text{Asp}_{\text{PFV}} > \text{voice} > \text{Asp}_{\text{RED}} > v$
 - reduplication can wait until additional material is inserted to be exponed (20)
 - pre-stress vowels are visible to infixation for AV but not PFV, (10) vs. (25)

open questions:

- can we reduce the alternation between AV $m-$ and $\langle m \rangle$ to a single infix location of $_V$?
 - is it feasible to say that pre-stress vowels are visible to AV and not PFV?
- how do we explain in /pm/ sequences the deletion of [p] in roots vs. [m] in causatives?
- does the affix ordering apply to right-aligned voice affixes, such as PV and LV?
 - this is unfortunately hard to test, but some Formosan languages do have prefixal PV
- what about the AV.FUT form $m-p-$ in Seediq? does AV $m-$ attach outside of FUT $p-$?
- what is the significance of voice and Asp_{RED} being ordered differently in Tagalog?
- TBD: the interaction between voice and mood (i.e. imperative and subjunctive forms)
 - the AV marker is consistently absent in AV.IMP forms across Formosan languages

Seediq

	AV	PV	LV	IV
PRES	m- or ⟨m⟩	-un	-an	s-
PFV	m-n- or ⟨m-n⟩	⟨n⟩	⟨n⟩ -an	-
PSV	m-RED- or RED⟨m⟩-	RED- -un	RED- -an	-
FUT	m-p-	RED- -un	RED- -an	-
IMP	∅	-i	-an-i	-
SUBJ	m- -a or ?	-o	-e	-

Puyuma

	AV	PV	LV	IV
PRES	m(a)/m(e)- or ⟨em⟩	-aw	-ay	-an-ay
FUT	RED-	RED- -aw	RED- -ay	RED- - an-ay
PROG	m(a)-RED- or RED⟨em⟩-	RED- -aw	RED- -ay	RED- - an-ay
IMP	∅	-u	-i	-an-i
IMP.NEG	RED-	RED- -u	RED- -i	RED- -an-i

PFV: =lra attaches outside of voice *and* pronominal clitics

Bunun

	AV	PV	LV	IV
PRES	m(a)-	-un	-an	is-
PFV	m(a)⟨in⟩-	⟨in⟩	⟨in⟩ -an	s⟨in⟩-
IMP	-a	-a-v	-a-v	-a-v

PFV: =in attaches outside of voice

FUT: na= attaches outside of voice

Atayal

	AV	PV	LV	IV
PRES	m- or ⟨m⟩	-un	-an	s-
PFV	m⟨in⟩- or ⟨in⟩	⟨in⟩	⟨in⟩ -an	?
FUT	p-	RED- -un	RED- -an	?
IMP	∅	-i	-i	-an-i
SUBJ	m- -a or ⟨m⟩ -a	-aw	-ay	-an-ay

Paiwan

	AV	PV	LV	IV
PRES	m- or ⟨em⟩	-en/in	-an	si-
PFV	na-m- or na- ⟨em⟩	⟨in⟩	⟨in⟩ -an	s⟨in⟩i-
IMP	-u/i	-u/i	-u/i	-an
SUBJ	-	-av	-av	si- -an

COMP: =anga attaches outside of voice

CONT: =anan attaches outside of voice

Tsou

	AV	PV	LV	IV
PRES	m(i)- or ⟨m⟩	-a	-o	-(n)eni

auxiliaries: AV: mi, mo, moh, moso

NAV: i, oh, os

IRR: te, ta, tena

IRR.NEG: 'ote (AV), teav'a (NAV)

PFV: =cu attaches to auxiliaries

HAB: =la attaches to auxiliaries

Kavalan

	AV	PV
PRES	m(e)- or ⟨em⟩	ma- or -an
IMP	-ka	-i-ka

PFV: u-, ni-, =ti attach outside of voice

FUT: qa- and =pa attach outside of voice

Amis

	AV	PV	LV	IV
PRES	ma/mi- or ⟨om⟩	ma- or -en	-an	sa-
IMP	∅	-en	-	sa- -en

PFV: =to attaches outside of voice

IPFV: =ho attaches outside of voice

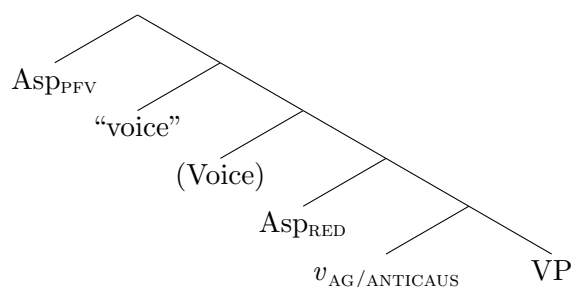
addendum

I use the term “voice” in this handout as a purely descriptive term, as is common in Austronesian literature, not to be conflated with Voice in the sense of the external argument introducing head

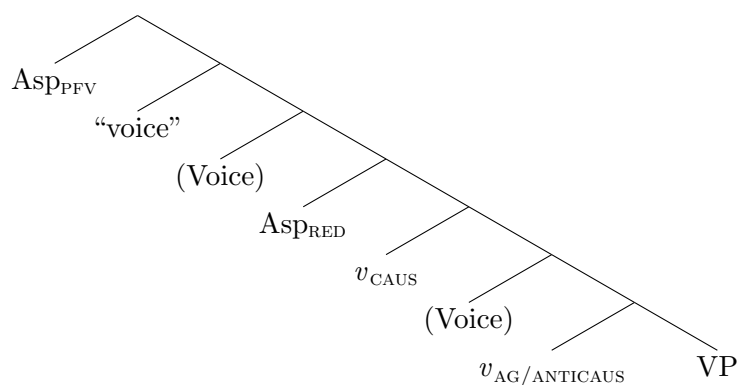
- it is a set of affixes that mark the verb and correspond with a “pivot” of the clause
- this pivot is a privileged argument for sentence position, extraction, case marking, etc.

Here are some (tentative) assumptions about the structure of the aspectual and verbal domain:

(27) intransitive and transitive verbs



(28) productive causative verbs



- “voice” and Voice could be the same entity, particularly in basic verbs like (1), but this could run into trouble if we assume causative constructions have two Voice projections like (2)
 - “voice” morphology only shows up once in productive causatives
 - could this be resolved with an applicative analysis of causatives (cite)?
 - “voice” morphology does show up twice with restructuring and other complementation
- the status of v_{ANTICAUS} is still unclear
 - there is no data showing the ordering relationship between Asp_{RED} and v_{ANTICAUS}
 - is anti-causative morphology hosted in Voice (cf. $\text{Voice}_{[-D]}$ in Schafer 2007) or in v ?
 - is it compatible with v_{CAUS} ? what is its relationship with stative marking?
- does Asp_{RED} only show up once in productive causatives (and restructuring)?